Formal Effects of Informal Labor and Work Permits Evidence from Venezuelan Refugees in Colombia

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Abstract

We analyze the Venezuelan refugee crisis in Colombia to separate the labor market effects of informal immigration and work permits. Using Synthetic Instrumental Variables, we find informal immigration displaced natives from both informal and formal sectors, implying high substitutability between worker types (elasticity ≈ 10). A triple difference in differences shows work permits reduced competition in the informal sector and raised it in the formal sector. Our model implies this created 23,100 formal jobs and roughly $42 million in annual tax revenue, which provides an economic rationale for granting refugees the right to work.

JEL Classification: D22, J15, J21, J46, J61.

Keywords: Immigration, Refugee crises, Work permits, Informality

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1 Introduction

Forced immigration episodes have long served as quasi-experimental settings to study the effect of immigration on labor markets, from the Mariel Boatlift (Card, 1990) and the Algerian war of independence (Hunt, 1992) to the Yugoslav wars (Angrist and Kugler, 2003) and the Syrian refugee crisis (Del Carpio and Wagner, 2015). However, these episodes typically bundle two distinct shocks: the arrival of immigrants and subsequent government policies, such as providing refugees with the right to work, that facilitate or restrict their integration.[1] The overall effect of immigration on the host country's labor markets likely depends on which rights are granted. Yet because the two shocks unfold simultaneously, existing research has little choice but to conflate the effects of policy interventions with those of the initial arrival. In this paper, we develop a theoretical framework that formalizes the impact of immigrants with and without work permits and provide empirical evidence of these mechanisms in the context of Venezuelan migration to Colombia, where we disentangle the effects of the inflow of immigrants and the regularization that followed on the labor market.

Colombia is an ideal setting to study the separate effects of immigration and work permit policies because, among episodes of large-scale irregular migration, it is among the few countries to have received millions of migrants without work authorization and only later granted them work permits en masse. By 2023, Colombia hosted 2.5 million forcibly displaced Venezuelans, one of the largest refugee populations in the world.[2] Venezuelans began arriving in the mid-2010s, mostly without a regular status that would allow them to work in the formal sector. Their numbers remained negligible until mid-2016. Over the course of several waves beginning in 2017, Colombia provided Venezuelans with a migratory status that included work permits. The gradual provision of these work permits provides quasi-experimental variation that allows us to identify the labor market effects of both (1) the predominantly informal labor supply shock caused by the arrival of Venezuelan migrants and (2) the subsequent provision of work permits.

We start by proposing an equilibrium model with three key features of labor markets in

[1] Work rights have accompanied arrival, or followed it, in several recent episodes: Ukrainian refugees received work authorization almost immediately across the EU (European Commission, 2022), Syrians in Turkey gained restricted, employer-tied permits with very low take-up, the Jordan Compact of 2016 and Kenya's 2021 Refugee Act expanded work rights for populations already hosted, and Spain regularized some 600,000 non-EU immigrants (Elias et al., 2025).

[2] Throughout the paper we refer to displaced Venezuelans in Colombia as migrants. UNHCR designates them as "refugees and migrants from Venezuela," and our title follows the literature on this episode, but most have not been granted refugee status under the 1951 Convention. The population combines refugees, irregular migrants and other displaced persons. We retain "the Venezuelan refugee crisis" as the conventional name of the episode. We take no stance on legal status, and use the precise terms (documented, undocumented) where it bears on the analysis, as with permit eligibility. Our framework applies wherever migrants initially lack, and later gain, formal work authorization.

developing economies: firms can use both informal and formal labor in production; natives can work for firms in salaried positions or in non-salaried work, meaning self-employment and other own-account arrangements rather than wage employment by a firm (nearly sixty percent of all workers in our data hold non-salaried positions); and natives choose between the informal and formal sectors based on wages and sector amenities, with a mobility parameter governing how strongly they re-sort when relative wages change. In this model, an informal labor supply shock, such as the arrival of immigrants without legal status, necessarily reduces natives' wages and salaried employment in the informal sector. Natives who can transition to non-salaried positions without substantial income losses choose to do so, and this transition both dampens the decline in informal wages and amplifies native displacement in salaried employment. Informal immigration's effect on native formal employment balances two forces: increased productivity from Q-complementarity between informal and formal workers, and increased competition against formal workers under diminishing returns to labor. Firms' demand for formal labor falls when informal and formal workers are gross substitutes and rises when they are gross complements. Even when worker types are gross substitutes, whether native formal employment falls on net depends on worker mobility, since displaced natives who re-sort into formal work offset the firms' substitution. The estimated sign of the formal-employment response therefore does double duty: it tests the model and locates the economy on the mobility spectrum.

In this framework, work permits shift some migrant labor supply from the informal to the formal sector, so natives' wages and salaried employment in the informal sector unambiguously rise. The effect on native formal workers is not signed by theory: migrant entry lowers formal wages, while the higher informal wage induces firms to demand more formal labor. Which force dominates determines whether permits create losers among native workers. The model does sign the aggregate: when informal and formal workers are gross substitutes, total formal employment rises despite any displacement of native formal workers.

Next, we empirically test the model's implications using data from the Colombian household surveys. We face three challenges to identification. First, immigrants choose where to locate based on economic conditions, leading to the standard endogeneity problem. Second, even departments that receive immigrants for plausibly exogenous reasons, such as proximity to the border, may follow different economic trajectories to departments that don't receive them. Third, departments with more immigrants tend to receive more work permits, creating spatial correlation between the two. To address these challenges, we employ two distinct research designs that separately identify the labor market effects of (1) the predominantly informal labor supply shock and (2) granting work permits to irregular migrants.[3]

[3]We characterize this as a "predominantly informal" labor supply shock because the inflow also included

To study the effects of the predominantly informal labor supply shock, we employ a shift-share design where travel distance between Colombian and Venezuelan cities operates as the share and the total number of migrants in Colombia in a given year acts as the shift. We further relax the share-exogeneity assumption embedded in our design (Goldsmith-Pinkham et al., 2020) by using the Synthetic Instrumental Variables method (SIV) (Gulek and Vives-i Bastida, 2023). This method uses synthetic controls to account for the empirical fact that border departments follow different economic trajectories to other departments.

We find that informal immigration displaced native salaried workers in both the informal and formal sectors. A 1 percentage point (pp) increase in the migrant-to-native ratio decreases native informal salaried employment by 0.27 pp and formal salaried employment by 0.14 pp. Based on average values at baseline, they translate to about a 3% drop in informal salaried employment and about a 1.4% drop in formal salaried employment.[4] The former result is predicted by the model, while the latter suggests that (1) native workers have limited mobility across the informal and formal sectors and (2) informal and formal labor are highly substitutable in production, with an estimated elasticity of substitution of approximately 10. Such a large elasticity implies that policies that make formal labor relatively cheaper (or informal labor relatively more expensive) can substantially incentivize firms toward greater formality. Moreover, we find that the immigration shock *increases* non-salaried employment, especially for men without high-school degrees, as predicted by the model. Total employment shows no significant change in the aggregate.

Subsequently we study the effects of work permits provided by the government to the irregular immigrants on Colombian workers. To do so we exploit the empirical fact that work permits primarily enable skilled immigrant men to transition to the formal sector, while having no sustained effect on other groups. This additional margin of comparison allows us to employ a triple difference-in-differences design that partials out the department-time specific immigration shocks identified earlier.

Our empirical results show that work permits cause natives to gain informal jobs and to lose formal jobs. Our model rationalizes these effects as arising from reduced competition in the informal sector and increased competition in the formal sector. We then calibrate the model with our empirical estimates and moments from the data and use it to quantify work permits' effect on formal job creation. We estimate that all work permits issued during our study period enabled 68,000 Venezuelans to transition from informal to formal jobs, creating

Colombian return-migrants, who retain legal access to formal employment. Our treatment excludes them by construction: its numerator counts only Venezuelan-born individuals. The distance instrument predicts only an early, one-time rise in their settlement that never tracks the sustained post-2017 buildup identifying our estimates (Appendix Figure H.9).

[4] Appendix Figure C.1 reports the effects in percentage terms for all skill groups.

a net increase of 23,100 formal jobs in the economy. Under the assumption that these new jobs paid the mean formal-sector wage, we estimate that granting work permits generated approximately 42 million USD in annual tax revenue.[5]

Our findings contribute to several strands of literature, which Appendix I discusses in more detail. First, this paper comprehensively reexamines labor market responses to immigration shocks (e.g., Card, 2009; Glitz, 2012; Borjas and Monras, 2017; Llull, 2018; Peri and Yasenov, 2019; Lebow, 2024).[6] The primary contribution of our paper is to separate the empirical estimates of the effects of the inflow of immigrants and those of subsequent policies aiming to integrate them into the labor markets. We show that while the informal immigration shock displaces natives in both the informal and formal sectors, granting work permits improves natives' prospects in the informal sector and increases the total number of formal jobs in the economy.

Our paper is closely related to the literature on Syrian refugees in Turkish labor markets, where studies using different identification strategies reached conflicting results (Del Carpio and Wagner, 2015; Erten and Keskin, 2021). As Gulek (2023) shows in his review of this literature, these contradictions stem from misinterpreting differential pre-trends between host and non-host regions as causal effects and from overlooking men's transitions to non-salaried positions. Adjusting for both, he finds that informal Syrian refugees decreased both informal and formal salaried employment of natives, with low-skill men transitioning to non-salaried work, and estimates an elasticity of substitution between informal and formal labor of approximately 10. Our Colombian estimates match on both counts. The key difference is that Turkey never granted Syrian refugees extensive work permits, so Gulek (2023) could only predict work permit effects as a counterfactual. Our setting tests these predictions with reduced-form evidence.

Second, we complement the literature on the dynamics between informal and formal labor markets.[7] Our framework unifies three modeling ingredients that this literature developed separately. From Ulyssea (2018) we take the demand side: firms jointly employ informal and formal labor and face an increasing expected cost of informal hiring. From Meghir et al. (2015) we take worker mobility, replacing search frictions with the discrete-choice labor supply framework of Card et al. (2018), which keeps the equilibrium in closed form and reduces worker mobility to a single parameter that our estimates can locate between

[5]Work permits may also improve the match between migrants and employers. Appendix G presents suggestive evidence. Wages of more exposed natives rise more in industries that are more formal and more skill-intensive. This pattern is consistent with skilled migrants moving into industries where they raise productivity beyond their labor supply.

[6]For a recent review of this literature, see Verme and Schuettler (2021).

[7]See Ulyssea (2020) for a review.

the segmented-markets and frictionless-Roy benchmarks. From Gulek (2023) we take the distinction between salaried and non-salaried work, which lets displaced salaried workers absorb shocks through self-employment rather than unemployment.

Third, our results on work permits build on the literature on formalization policies in developing countries (Monteiro and Assunção, 2012; De Andrade et al., 2016; Rocha et al., 2018) and developed countries (Elias et al., 2025).[8] Most similar to our setting, Bahar et al. (2021) study the effects of granting Venezuelan migrants work permits and find negative but negligible effects on the formal employment rate of Colombian workers. Our results differ for two reasons. Economically, Bahar et al. (2021) do not distinguish between salaried and non-salaried positions. Econometrically, they use a standard difference-in-differences design, which we show confounds the effects of immigration and work permits. We partial out the department-level impact of immigration with a triple difference. Also studying PEP, Ibáñez et al. (2025) use a fuzzy regression discontinuity at the registration cutoff of a 2018 migrant census and find large gains for migrants themselves in consumption, labor income and health. We complement their results by showing that permits also reallocate native labor and create formal jobs on net.

Finally, our work fits in with several other studies that examine the impact of the Venezuelan migration crisis on Colombia's labor market. Early studies found little to no employment impact (Santamaria, 2020; Caruso et al., 2021). These studies rely on employment measures that pool salaried and non-salaried work (the aggregation that, as we show, masks displacement) and do not correct for the differential trends between border and interior departments. The two papers with the most comprehensive data and identification are Delgado-Prieto (2024) and Lebow (2022). Delgado-Prieto (2024) uses distance and historical settlement patterns as instruments and finds a decline in formal native employment and in informal salaried employment. Lebow (2022) uses a historical migration instrument and finds significant wage effects but little to no impact on employment. We show that natives lose salaried jobs in both the informal and formal sectors across all skill groups. Two factors account for these differences. First, since border departments follow different labor market trajectories than interior departments, the share-exogeneity assumption embedded in standard spatial IV-DiD designs (Goldsmith-Pinkham et al., 2020) likely fails. While existing work controls for trends parametrically (Delgado-Prieto, 2024; Lebow, 2022) or applies standard synthetic controls (Penaloza-Pacheco, 2022), our SIV estimator jointly adjusts for differential pre-trends nonparametrically and instruments for endogenous treatment. Second, we estimate the salaried and non-salaried margins separately for each skill-gender group. Both papers report decompositions by employment type, but pooled across demographic groups. Our

[8] See Clemens et al. (2018) for why granting refugees work permits can benefit both refugees and natives.

group-level estimates reveal the offsetting movements the aggregates mask: losses in both informal and formal salaried work across skill groups, with low-skill men transitioning into self-employment.

2 Background and Data

2.1 The Venezuelan Refugee Crisis in Colombia

Venezuela began to experience out-migration in the early 2000's in response to President Chavez's failed political and economic reforms. Following his death in 2013, the economic and political crisis deepened. Venezuelans had been arriving in Colombia in small numbers for years, but mass migration began in late 2015, and their numbers grew rapidly from mid-2016. Figure 1a shows how the number of Venezuelan migrants in Colombia has evolved over time. In 2022, Venezuelans made up approximately 5% of the Colombian population.

Figure 1b maps the migrant-to-native ratio (Venezuelans per native) across departments, our unit of analysis, in 2019. Venezuelans are more densely located in departments closer to the border (Caruso et al., 2021). The distance from the populous states of Venezuela predicts the migrant-to-native ratio across departments, which constitutes one of the key building blocks of our identification strategy.

In response to the inflow, the Colombian government established a temporary visa program called *Permiso Especial de Permanencia* (PEP), which granted Venezuelans authorization for formal employment and access to social security benefits. Appendix B documents the program in detail. PEP was implemented in three waves, which we refer to as PEP-1, PEP-2 and PEP-3. The first two, launched in July 2017 and February 2018, required proper documentation and a legally recognized migratory status. Together they issued nearly 182,000 permits. Because obtaining a Venezuelan passport was prohibitively costly, these early waves exhibited strong positive selection, with eligibility predominantly restricted to highly skilled migrants.

The third and largest wave was the byproduct of RAMV, a nationwide census of undocumented, hence PEP-1 & 2 ineligible, Venezuelan migrants conducted between April and June 2018. Registration was not explicitly or implicitly related to the possibility of obtaining formal migratory status, and RAMV identified 442,462 undocumented Venezuelans. On July 25, 2018, just days before his term ended, President Juan Manuel Santos unexpectedly issued a decree allowing everyone listed in the RAMV to apply for a new round of PEP. Sixty-four percent, or about 280,000, received one.

The differential documentation requirements between PEP waves 1 and 2 (formal docu-

mentation required) and PEP-3 (no documentation required) resulted in Venezuelans with varying skill levels accessing formal employment at different points in time. Figure 2 plots formal employment of Venezuelans in six education-gender categories, relative to its level in 2017q3, when the rollout of PEP-1 began.[9] We define three skill groups by education: less than high school (low-skill), high school but not college (middle-skill), and college or above (high-skill). Each is split by gender. Pre-PEP-1 the series are flat: at most one of six quarters per group differs from the baseline, always below it, as expected when the population base is growing. Formal employment then rises for the two treated groups in the order the rollout predicts. High-skill men, eligible under PEP-1 and PEP-2, first become significant at the 1% level in 2018q3. Middle-skill men, who in practice gained access only under PEP-3, first do so in 2019q1. Both stay significant through 2019q4. These onsets survive a sharpened false-discovery-rate correction for multiple testing across the six groups (Benjamini et al., 2006; Anderson, 2008), and Appendix B reports two alternative corrections. Both onsets lag the wave that made the group eligible, as the program's mechanics imply: application windows closed one to two quarters after each decree, and approval and job search added further weeks (Appendix B). High-skill men's onset therefore coincides with the PEP-3 decree, but PEP-3 does not produce it: they were eligible from PEP-1, and their point estimates climb steadily from that wave until the cumulative gain clears the threshold in 2018q3. No control group shows a sustained response: low-skill men and women and high-skill women never clear the 1% threshold, and middle-skill women clear it in a single quarter. High-skill women's series rises modestly from 2018q3 onward but survives no multiple-testing correction. We treat high-skill women as untreated and show in Section 5.3 that our estimates are insensitive to this choice.

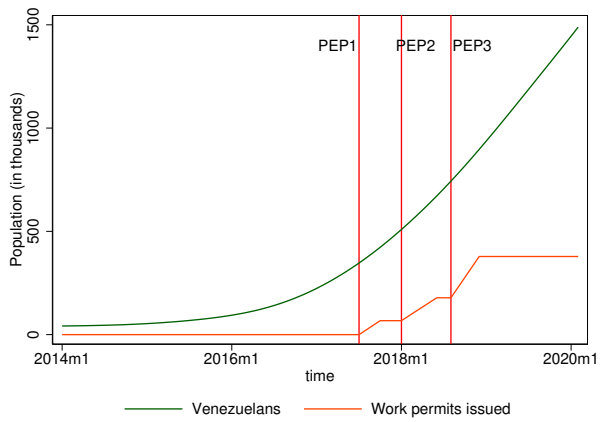
In summary, the Venezuelan refugee crisis in Colombia began as an informal labor supply shock, but the subsequent granting of work permits moved high-skill and then middle-skill Venezuelan men into formal employment.

2.2 Labor market statistics

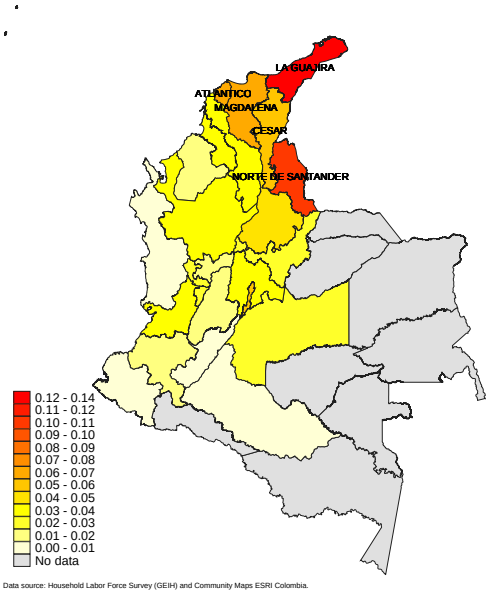
Information on informal and formal labor market outcomes for both Venezuelan and native workers is derived from the 2007–2019 Colombian household labor force survey (*Gran Encuesta Integrada de Hogares*, GEIH) conducted by the Colombian National Statistics De-

[9] The series are normalized to 2017q3, so the levels at that quarter are not visible. They total about 26,000 formally employed Venezuelans: 10,100 high-skill women, 6,500 high-skill men, 4,300 low-skill men, 3,800 middle-skill men, 1,600 middle-skill women and no low-skill women. Formal employment requires work authorization, so this pre-policy stock consists of migrants who already held regular status; the PEP waves acted on the irregular population.

Figure 1: The Venezuelan Refugee Crisis in Colombia



(a) Immigration and Work permits



(b) Cross-sectional distribution of the migrant-to-native ratio

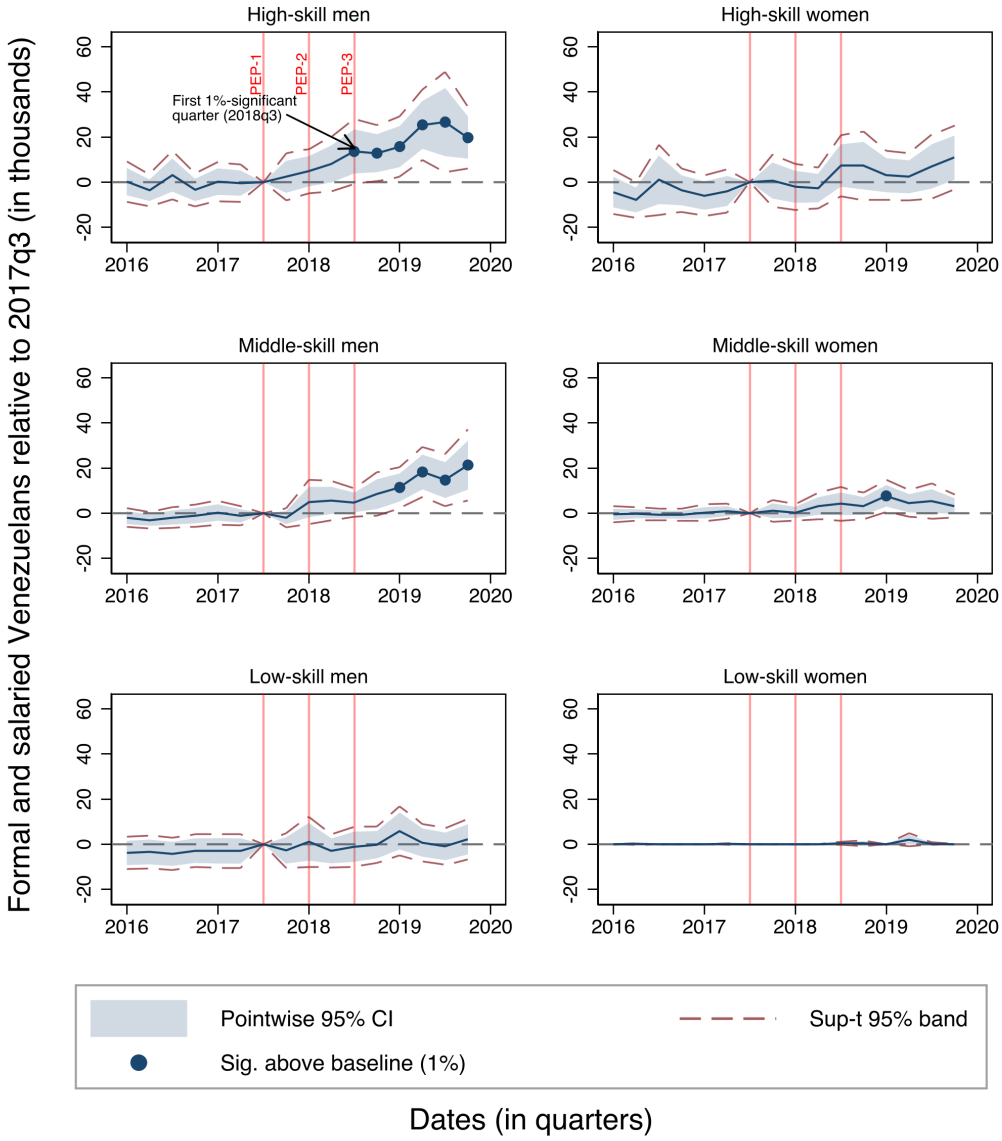
Notes: Author's calculations using GEIH and administrative permit records. Panel 1a plots the LOWESS-smoothed number of Venezuelan-born residents aged 18–65 in the 24 GEIH departments (green line) and the cumulative number of permits issued there (orange line), counting PEP-3 permits for ages 18–65 only. Panel 1b maps the migrant-to-native ratio in each of the 24 GEIH departments in 2019.

partment (*Departamento Administrativo Nacional de Estadística*, DANE).[10] GEIH surveys approximately 20,000 households monthly and maintains representativeness at the department level across 24 of Colombia's 33 departments. Since 2013, GEIH has included a migration module recording each respondent's country of birth, together with citizenship and place of residence one and five years prior to the survey date. "Natives" are all residents not born in Venezuela. Section 4 also excludes Colombian return-migrants.

All regression analyses aggregate the individual-level GEIH microdata to cells using survey weights. For the migration-shock analysis of Section 4, we collapse the data to department-year cells: each outcome is the survey-weighted mean of the corresponding individual indicator (for example, informal salaried employment) among working-age individuals in the relevant demographic group, yielding a balanced panel of 24 departments over 13 years ($N = 312$). For the work-permit analysis of Section 5, we collapse to department × skill-sex group × quarter cells (24 departments × 6 groups). Throughout the paper, the subscript r indexes these 24 departments. The note to every figure and table that reports an estimate states the corresponding number of observations. The migrant-to-native ratio that measures

[10] This dataset is publicly available on DANE's website.

Figure 2: Venezuelan formal employment relative to the onset of PEP-1, by skill-sex group



Notes: Author's calculations using GEIH. Each series plots the group's survey-weighted total of formally employed, private salaried Venezuelans in each quarter, minus its 2017q3 total, the quarter PEP-1 began (the horizontal dashed line at zero). Shaded areas are pointwise 95% confidence intervals of the difference, computed from the joint variance of the survey-weighted quarterly totals, with standard errors clustered at the household-by-survey-month level; the two dashed lines bracketing each series are simultaneous 95% sup-t confidence bands (Montiel Olea and Plagborg-Møller, 2019), which jointly cover each series' entire path. The three vertical dashed lines mark PEP-1 (2017q3), PEP-2 (2018q1), and PEP-3 (2018q3). Filled markers denote quarters significantly above the baseline at the 1% level; every such quarter also survives a sharpened false-discovery-rate correction for multiple testing across the six groups (Benjamini et al., 2006; Anderson, 2008). The sample consists of Venezuelan-born individuals aged 15–64: $N = 36{,}711$ individuals in 18,717 household-by-survey-month clusters.

the migration shock is built inside the GEIH from these same cells: for each department and year we divide the survey-weighted count of Venezuelan-born residents by the count of Colombian-born residents. No external migration-flow series is merged in, so the treatment and the outcomes come from a single household survey and are aggregated identically. External data enter the main design in one place only, the time-invariant distance instrument of Section 4, whose origin weights come from the administrative registry of Venezuelan migrants. Administrative permit records enter only in permit counts and Appendix G.

We construct three employment categories: informal salaried, formal salaried, and non-salaried. We exclude government employment. Salaried workers are employed by private companies for regular wages. Non-salaried workers encompass all other arrangements, primarily self-employment. The distinction matters because a native who loses an informal salaried job and moves into self-employment remains "employed" in the GEIH, so aggregate employment rates can stay flat while firms replace natives with migrants at scale. We therefore analyze the salaried and non-salaried margins separately, and do not split non-salaried work by formality, since the overwhelming majority operate informally. Employment rates across the three categories differ sharply by gender and education (Appendix Figure A.1). Salaried employment rises with education and is higher for men. Non-salaried work accounts for 58 percent of all employment, but for 71 percent among low-skill men and 37 percent among high-skill men (Appendix Table A.2). Appendix Figure C.9 reports the corresponding distribution of employment types for Venezuelan migrants.[11]

We classify employment as *formal* or *informal* based on self-reported health and pension coverage, to which Colombian employers must by law contribute on their workers' behalf. Workers have no incentive to misreport coverage, since it is illegal to employ informally but not to work informally. The resulting informality rates are also consistent with the literature (Ulyssea, 2020): informality decreases with education and is higher in less developed departments and industries, such as agriculture, known to rely on informal labor.

Appendix Figures A.1b and A.1c show the formality rate of salaried workers across select industries and firm sizes. Although there is significant heterogeneity, formal and informal workers coexist everywhere. The formality rate across sectors ranges from 24% in private household work to 91% in financial intermediation. Moreover, formality rate increases as firms get bigger: ranging from 14% in firms with 2 to 3 employees, to 93% in firms with more than 101 employees. This relationship can be rationalized by the fact that larger firms are more visible and therefore more likely to be punished if hiring informally (Ulyssea, 2020).

[11] Appendix Table A.1 reports income and wages.

3 Model

We develop a framework that formalizes how two shocks, the arrival of migrants who can only work informally and the subsequent granting of work permits, affect native labor market outcomes in an economy with a large informal sector. The framework has two building blocks. On the demand side, a representative firm employs informal and formal salaried labor in production and faces an increasing expected cost of informal hiring. On the supply side, workers make two nested choices: which sector to work in, and how many hours to supply within it, where informal workers further divide their hours between salaried work and non-salaried self-employment. To keep the exposition transparent, the model abstracts from skill heterogeneity across workers; we discuss how the predictions differ across worker groups where that heterogeneity sharpens the empirical tests. Appendix E develops two extensions of the model, one with two skill types and one with heterogeneous firms, and shows that the predictions below survive.

3.1 Setup

Labor Demand

A representative firm employs informal and formal salaried workers as the sole factors of production. Following Ulyssea (2018), we assume that while formal workers impose a constant marginal cost, hiring informal workers entails an increasing and convex expected cost, reflecting the greater enforcement risk faced by larger informal operations (De Paula and Scheinkman, 2011). This cost structure generates the negative correlation between firm size and informal employment documented in Section 2.2. The firm maximizes profits by choosing employment of both types:

$$\max_{\ell_i,\ell_f} \; F(\ell_i,\ell_f) - \tau(\ell_i)\, w_i - w_f\, \ell_f, \qquad F(\ell_i,\ell_f) = \theta\big(\eta\ell_i^{\rho} + (1-\eta)\ell_f^{\rho}\big)^{\frac{\alpha}{\rho}}, \tag{1}$$

where w_i and w_f denote informal and formal wages, $\tau(\ell_i)$ captures the convex expected cost of hiring ℓ_i informal workers with $\tau' > 0$ and $\tau'' > 0$, θ is Hicks-neutral productivity, $0 < \alpha < 1$ imposes decreasing returns to scale appropriate for short-run adjustments, $\sigma = \frac{1}{1-\rho}$ is the elasticity of substitution between informal and formal labor, and η governs the relative productivity weights.[12]

[12] For analytical simplicity, we interpret w_f as the gross formal wage, abstracting from payroll taxes.

Labor Supply: Hours

Conditional on a sector, workers choose hours. A formal worker solves

$$V_f(w_f) = \max_{h_f} \; w_f h_f - \frac{\delta_f}{1+\delta_f} h_f^{\frac{1+\delta_f}{\delta_f}}, \tag{2}$$

which yields the labor supply function $h_f = w_f^{\delta_f}$, with δ_f the hours elasticity.

Informal workers face a richer choice set, as they can work in salaried positions for the representative firm or in non-salaried activities.[13] Drawing on the home production literature (Gronau, 1977), we model non-salaried work as generating income $g(h_{ns})$, where $g(\cdot)$ is increasing and strictly concave ($g' > 0$, $g'' < 0$), capturing diminishing returns to self-employment. The informal worker solves

$$V_i(w_i) = \max_{h_i, h_{ns}} \; w_i h_i + g(h_{ns}) - \frac{\delta_i}{1+\delta_i} (h_i + h_{ns})^{\frac{1+\delta_i}{\delta_i}}. \tag{3}$$

The optimality condition $g'(h_{ns}) = w_i$ states that workers allocate time to non-salaried activities until the marginal return equals the informal wage. When competition from migrants reduces w_i, workers substitute toward non-salaried work, with the magnitude of the transition inversely related to the curvature of $g(\cdot)$.[14]

Labor Supply: Sector Choice

Each native draws a permanent pair of sector tastes and works in the sector with the higher total payoff, following the discrete-choice labor supply framework of Card et al. (2018). Worker n's payoff from sector $s \in \{i, f\}$ is $\ln V_s + \lambda(\mu_s + \epsilon_{sn})$, where the taste draws ϵ_{sn} are independent standard Type-I extreme value across workers and sectors, μ_s is a common amenity shifter, and $\lambda > 0$ converts amenities into consumption-equivalent units. Tastes capture how workers value the non-market attributes of each sector (e.g., pension and health coverage, schedule flexibility, etc.) and differ across workers.

Worker n selects the sector with the higher payoff. She chooses formal work when

$$m \ln V_f + \mu + \epsilon_{fn} \;\geq\; m \ln V_i + \epsilon_{in}, \qquad m \equiv \frac{1}{\lambda} \geq 0, \tag{4}$$

[13] The dichotomy in which informal workers hold salaried and non-salaried positions while formal workers hold only salaried positions is a close approximation of the Colombian data, in which non-salaried work is overwhelmingly informal.

[14] The model has no search frictions and hence no involuntary unemployment: workers displaced from salaried jobs adjust through the non-salaried margin. Consistent with this, we find no effect of the migration shock on native unemployment or labor force participation (Appendix Figure C.6).

where $\mu \equiv \mu_f - \mu_i$ is the mean non-wage difference between the two sectors, which can favor either sector, and m is the *mobility* parameter. Below, μ takes different values for natives and migrants, and for migrants with and without work permits. The extreme-value distribution delivers the logit share of natives in formal work, $P_f = \frac{e^{\mu + m \ln V_f}}{e^{\mu + m \ln V_f} + e^{m \ln V_i}}$: μ pins down the baseline formal share, and m governs how strongly natives re-sort when relative sector values change.

The parameter m spans the informality debate (Magnac, 1991; Maloney, 2004; Perry et al., 2007).[15] At $m = 0$, taste dispersion is so large that sector shares do not respond to wages: each worker is effectively endowed with the sector her taste draw assigns, which is the segmented-markets benchmark common in the literature. As $m \to \infty$, tastes vanish and workers sort frictionlessly on wages, as in a textbook Roy model.[16] Intermediate values interpolate between informality as exclusion and informality as exit. Rather than imposing a position on this spectrum, we derive which predictions depend on it and let the estimates locate the economy.

Migrants

Migrants supply salaried labor inelastically.[17] Within a market, migrant and native labor are perfect substitutes.[18] Without a work permit, formal employment is impossible, which we model as an infinite amenity penalty ($\mu^{mig} = -\infty$): the migrant formal share is zero due to law rather than preference. Work permits remove the prohibition, and permit holders then face the same discrete choice as natives, with their own finite amenity shifter μ^{mig}.[19]

[15] See Ulyssea (2020) for a review.

[16] For a search-theoretic treatment of worker flows between formal and informal jobs, see Meghir et al. (2015). Our static discrete-choice formulation keeps the equilibrium system tractable while preserving the reallocation margin their framework emphasizes.

[17] We abstract from migrants' non-salaried work. Non-salaried work is not a market in the model for any group: $g(\cdot)$ is concave in the worker's own hours, so no worker's return depends on any other worker's non-salaried activity. Giving migrants the same option would therefore not create competition with natives on that margin. Capturing that would require congestion in non-salaried work, which we model for no one. Since Venezuelans did enter self-employment in large numbers (Appendix Figure C.9), congestion would dampen natives' movement into non-salaried work, so our estimate of that margin is conservative.

[18] We assume migrants and natives are perfect substitutes within skill groups. Relaxing this toward the imperfect substitutability documented by Ottaviano and Peri (2012) would introduce complementarity between migrants and natives that raises natives' marginal product when migrants arrive, attenuating the displacement our framework predicts and making our estimates conservative. Across skill groups, the two-skill extension in Appendix E.9 allows imperfect substitution and leaves the predictions unchanged.

[19] Amenity values can differ between natives and migrants for several reasons; for example, migrants may value retirement contributions less than natives if they expect to return home or migrate onward in the future. We do not restrict μ^{mig}; the data discipline it.

Equilibrium

Wages clear the informal salaried and formal labor markets; non-salaried employment adjusts through worker optimization alone.[20] Log-linearizing around the initial equilibrium, native labor supply to the two salaried markets responds to both wages:

$$\begin{aligned} d\ln L_{i,s}^{S} &= \left(\varepsilon_{is} + mP_f\varphi_i\right) d\ln w_i \;-\; mP_f\varphi_f\, d\ln w_f, \\ d\ln L_{f}^{S} &= \left(\delta_f + m(1-P_f)\varphi_f\right) d\ln w_f \;-\; m(1-P_f)\varphi_i\, d\ln w_i, \end{aligned} \tag{5}$$

where ε_{is} denotes the informal salaried hours elasticity and $\varphi_s \equiv d\ln V_s / d\ln w_s$ measures how the value of sector s responds to its wage (Appendix E provides the expressions). The own-wage terms combine hours and entry. The cross-wage terms capture reallocation: a higher formal wage pulls informal natives out of informal work, at a rate that scales with mobility, and symmetrically for the formal market. Given migrant labor R and the migrant amenity shifter μ^{mig}, migrants select into the informal and formal sectors, supplying $R_i = (1-P_f^{mig})\,R$ and $R_f = P_f^{mig}\,R$, each measured relative to its own market's baseline size, where P_f^{mig} is the migrant counterpart of the formal share above.[21] Market clearing requires

$$\begin{aligned} \varepsilon^{D}_{\ell_i,w_i}\, d\ln w_i + \varepsilon^{D}_{\ell_i,w_f}\, d\ln w_f &= d\ln L^{S}_{i,s} + R_i, \\ \varepsilon^{D}_{\ell_f,w_i}\, d\ln w_i + \varepsilon^{D}_{\ell_f,w_f}\, d\ln w_f &= d\ln L^{S}_{f} + R_f, \end{aligned} \tag{6}$$

where $\varepsilon^{D}_{\ell_x,w_y}$ is the elasticity of labor demand for type x with respect to the wage in sector y. At $m = 0$ the cross-wage terms vanish and the two markets become fully segmented. Appendix E shows that, under a mild regularity condition that holds at our estimated parameters, the system has a unique solution for every $m \geq 0$.

3.2 The Effects of an Informal Labor Supply Shock

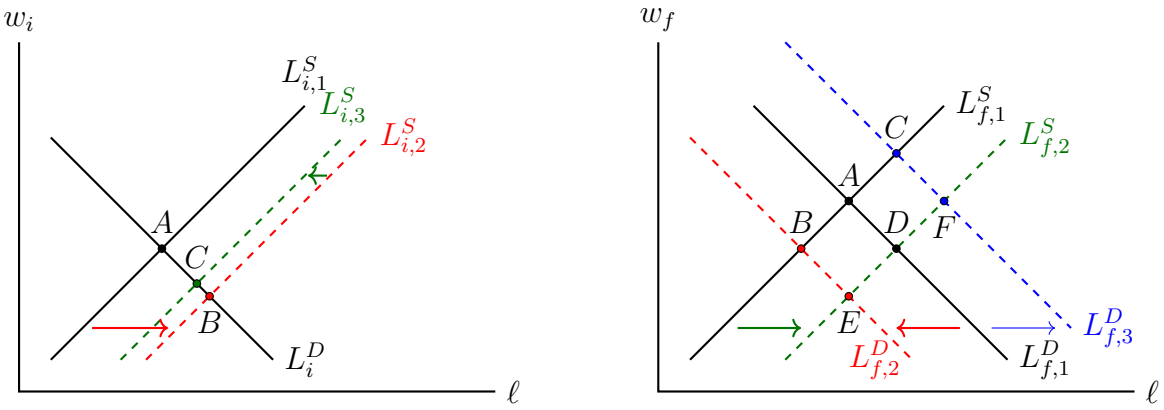
Migrants without permits supply only informal labor: $R_i = R > 0$ and $R_f = 0$. Panels (a) and (b) of Figure 3 show how the informal and formal labor market equilibria respond to this shock under different model parameters. The following propositions characterize the responses of native employment; all derivations and proofs are in Appendix E.

[20] Throughout, sector employment means labor input: the number of workers in a sector times their hours. The extensive margin across sectors is the formal share P_f, which responds only through mobility. In the baseline the non-salaried margin is an hours choice inside the informal worker's problem, so the workers with some non-salaried activity are the informal workers. Appendix E.13 re-derives the model with non-salaried work as a distinct occupation, where that margin is a count of workers, and the predictions are unchanged.

[21] We treat the shocks to migrant labor supply as the objects of interest and abstract from migrants' further re-sorting in response to the induced wage changes; Appendix E shows that allowing migrants to re-sort leaves the model's predictions unchanged.

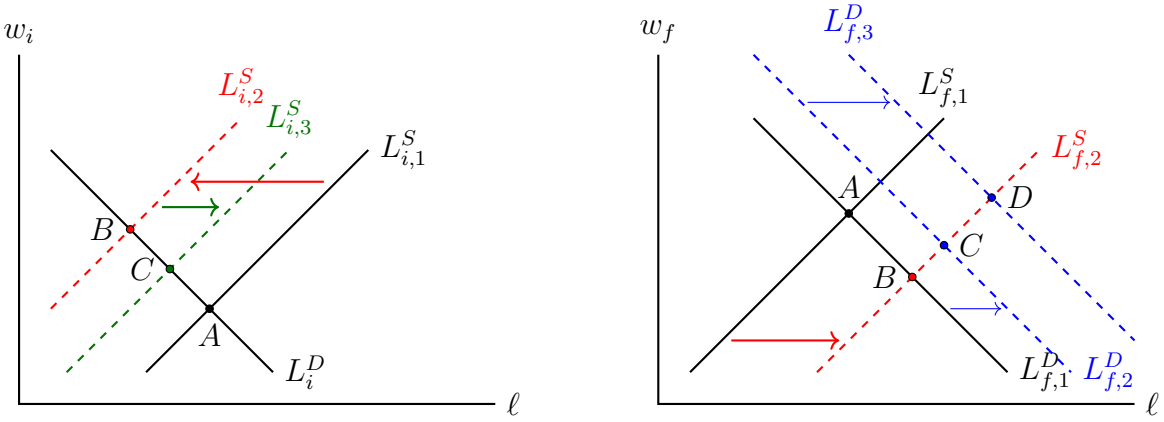
Figure 3: The Two Shocks in the Dual Labor Market

Shock 1: The informal labor supply shock



(a) Informal Sector (b) Formal Sector

Shock 2: Work permits (starting from the post-migration equilibrium)



(c) Informal Sector (d) Formal Sector

Notes: A stylized dual labor market responding to the two shocks. Color code: red marks migrant labor supply shifts, dark green native reallocation ($m > 0$), and blue firms' labor demand responses; $\alpha < \rho$ denotes gross substitutes and $\alpha > \rho$ gross complements between informal and formal labor. Panels (a) and (b) depict the informal labor supply shock. In Panel (a), the shock shifts informal supply from $L^S_{i,1}$ to $L^S_{i,2}$ (A to B); with mobility ($m > 0$), natives' exit toward the formal sector shifts supply partially back to $L^S_{i,3}$ (equilibrium C), attenuating the wage decline without reversing it. In Panel (b), the lower informal wage shifts formal labor demand to $L^D_{f,2}$ under substitutes and to $L^D_{f,3}$ under complements, while reallocation shifts native formal supply to $L^S_{f,2}$, by more the higher is m; reallocation alone would reach D. Any point in the region bounded by B, C, E, and F is a potential equilibrium; under substitutes it moves from B (no mobility) toward E as mobility rises. It reaches E, where the supply shift offsets the demand shift and native formal employment returns to baseline, at $m = m^*$ (Proposition 2). Panels (c) and (d) depict the work-permit shock, starting from the post-migration equilibrium A. In Panel (c), formalizing migrants leave informal work, shifting supply from $L^S_{i,1}$ to $L^S_{i,2}$ (A to B) and raising the informal wage; with mobility, natives drawn back into informal work shift supply to $L^S_{i,3}$ (equilibrium C), moderating the increase. In Panel (d), migrant entry shifts formal supply from $L^S_{f,1}$ to $L^S_{f,2}$ (B absent any demand response). Under substitutes the higher informal wage raises formal labor demand: a modest response ($L^D_{f,2}$) yields C, where the formal wage falls, and a strong response ($L^D_{f,3}$) yields D, where it rises. Total formal employment expands in every case; whether native formal workers lose or gain is an empirical question.

In the informal sector (Panel a), the shock shifts labor supply from $L^S_{i,1}$ to $L^S_{i,2} \equiv L^S_{i,1} + R_i$. Without mobility ($m = 0$), the equilibrium moves from A to B: the informal wage falls and natives lose informal salaried jobs. With mobility ($m > 0$), some natives respond to the lower informal wage by moving to the formal sector, which shifts supply partially back to $L^S_{i,3}$ (equilibrium C). This reallocation is second order in the informal sector: it moderates the decline in wages, and natives lose informal wages and salaried employment either way.

Proposition 1. *In response to an informal labor supply shock, native informal salaried employment falls for every level of substitutability between informal and formal labor* $\sigma = \frac{1}{1-\rho}$ *and every mobility level* $m \geq 0$.

In the formal sector (Panel b), the responses of both firms and workers matter. On the demand side, more informal employment has two competing effects: it makes formal workers more productive because of Q-complementarity, and it also creates competition against formal employees through diminishing returns to labor. The two forces net out to the sign of $\alpha - \rho$: hiring informal labor raises formal workers' marginal product through the production complementarity and lowers it through decreasing returns.[22] When informal and formal workers are gross substitutes in production ($\alpha < \rho$), the lower informal wage incentivizes firms to replace formal employees with informal ones, shifting labor demand down from $L^D_{f,1}$ to $L^D_{f,2}$; when they are complements, labor demand shifts up to $L^D_{f,3}$. On the supply side, mobile natives transition toward formal work, pushing formal labor supply from $L^S_{f,1}$ toward $L^S_{f,2}$, by more the higher is mobility. When informal and formal workers are gross substitutes and mobility is zero, the equilibrium moves from A to B; with mobility it slides from B toward E, where employment returns to its level at A. Any point within the region bounded by B, C, E, and F is feasible: both employment and wages can move in either direction. Proposition 2 resolves the ambiguity into a threshold condition.

Proposition 2. *Suppose informal and formal labor are gross substitutes* ($\alpha < \rho$). *There exists a threshold* $m^* > 0$ *such that native formal salaried employment falls if* $m < m^*$ *and rises if* $m > m^*$.

At $m = 0$ native formal employment falls. As m rises, worker reallocation strengthens until it exactly offsets firm substitution at m^*. Beyond m^* native formal employment rises on net. Natives can move toward formal work even when the formal wage falls, because the

[22] Decreasing returns make the second force possible. Under constant returns ($\alpha = 1$), $\alpha - \rho = 1 - \rho > 0$ for every σ, so the two types are always gross complements. The substitutes case requires $\alpha < \rho$, that is $\sigma > 1/(1-\alpha)$. At our calibrated $\alpha = 0.49$ this means $\sigma > 1.96$. In the usual terminology two inputs are q-complements when the net effect $\alpha - \rho$ is positive, so q-complementarity in that sense coincides with gross complementarity. We use Q-complementarity for the first force alone, whose strength falls as ρ rises.

informal wage falls by more.[23] The threshold is increasing in the strength of the demand response and decreasing in the strength of reallocation; its closed form is in Appendix E. A decline in formal employment therefore carries information: it implies that firms' ability to substitute informal for formal workers is a stronger force than natives' ability to move across sectors.

Proposition 3. *There exists a threshold* $m^{**} \in (0, \infty]$ *such that native non-salaried employment rises if* $m < m^{**}$ *and falls if* $m > m^{**}$.

The non-salaried margin does not appear in the figure, but it faces the same tension. The lower informal wage pushes informal workers toward self-employment through $g'(h_{ns}) = w_i$, while reallocation drains the pool of informal natives who supply that margin. For mobility below m^{**} the direct effect dominates and non-salaried employment rises.

Heterogeneous Effects Across Worker Groups

The framework generates heterogeneous predictions across worker subgroups that provide additional empirical tests. First, Colombia's binding minimum wage in the formal sector constrains wage adjustment for low-skilled workers. Formal labor demand shifts therefore translate into employment changes rather than wage changes for this group, implying larger percentage employment effects for low-skilled natives.

Second, the magnitude of the transition from salaried to non-salaried work depends on the curvature of the earnings function $g(\cdot)$. Appendix Table E.2 documents that this curvature varies systematically across demographic groups, being lower (implying greater responsiveness) for men and rising with education, so that responsiveness falls with education. These patterns predict that low-skilled men should exhibit the strongest transitions to non-salaried employment following informal wage declines.

3.3 The Effects of Work Permits

Work permits remove the legal prohibition on migrant formal employment: μ^{mig} jumps from $-\infty$ to a finite value, and permit holders whose taste draws favor formal work leave the informal sector. The migrant formal share rises from zero to P_f^{mig}, so a mass $r = P_f^{mig} R$ of migrants reallocates from informal to formal supply. Panels (c) and (d) of Figure 3 show how the two labor markets respond.

[23] At $m = m^*$ native formal employment is unchanged, but both wages are strictly lower and total informal labor input is strictly higher, so employment neutrality is not welfare neutrality: every native's sector value falls. Appendix Remarks E.5.1 and E.5.2 give the limits $m = 0$ and $m \to \infty$ in closed form and the argument at m^*.

In the informal sector (Panel c), starting from the post-migration equilibrium A, formalizing migrants leave informal work, shifting supply from $L^S_{i,1}$ to $L^S_{i,2}$.[24] Without mobility, the equilibrium moves from A to B: reduced competition raises the informal wage and natives gain informal salaried jobs. With mobility, some natives are drawn back into the now more attractive informal sector, shifting supply to $L^S_{i,3}$ (equilibrium C) and moderating the wage increase. Both forces, the higher wage and native re-entry, raise native informal employment.[25]

Proposition 4. *In response to work permits, native informal salaried employment rises for every substitutability level σ and every mobility level $m \geq 0$.*

In the formal sector (Panel d), two forces collide. Migrant entry shifts labor supply from $L^S_{f,1}$ to $L^S_{f,2}$, pushing the formal wage down. At the same time, the higher informal wage raises firms' demand for formal labor when informal and formal workers are gross substitutes, pushing the formal wage up. Which force dominates determines whether natives in the formal sector lose or gain. If migrant entry dominates (equilibrium C), the formal wage falls and natives lose formal jobs: permits benefit informal natives at the expense of formal natives. If the demand response dominates (equilibrium D), the formal wage rises and no native worker loses: informal natives benefit from reduced competition, and formal natives benefit from firms' increased demand for formal labor. The latter case requires the informal market to absorb the migrant exodus almost entirely through wages rather than employment: informal labor demand must be steep and natives' informal labor supply nearly inelastic, so that the departure of migrants raises the informal wage enough to overturn migrant entry in the formal market. Theory alone does not determine the sign. Whether work permits benefit all workers or create some losers among native workers is an empirical question, which Section 5 answers.

Proposition 5. *Suppose informal and formal labor are gross substitutes ($\alpha < \rho$). In response to work permits, total formal employment, natives plus formalized migrants, rises for every mobility level $m \geq 0$.*

Total formal employment is read off the formal labor demand curve at the new wages, and it expands in both cases: the higher informal wage raises firms' demand for formal labor, and formalized migrants fill the new positions. Mobility affects only the magnitude: as m rises,

[24] The panel plots total labor supply, including migrants. Total informal employment falls as migrants leave, while native informal employment rises: natives move up along their own supply curve as the wage increases.

[25] Formally, Proposition 4 requires the permit shock to raise the informal wage. At the estimated elasticities this holds unless the formal sector employs less than roughly two percent of salaried workers, far from the Colombian share of 62 percent; Appendix E states the exact condition.

natives flow back into the informal jobs migrants vacate, which damps the net formalization gain but never reverses it. Regularization also lowers the economy-wide informality rate for every degree of substitutability and, at our parameter values, at every mobility level. The more substitutable the two types are, the larger the fall (Appendix Corollary E.6.1).

3.4 Testable Predictions and Empirical Guidance

The framework yields five testable predictions. An informal labor supply shock (1) lowers native informal salaried employment, (2) lowers native formal salaried employment when substitutability is high and mobility is low, and (3) moves natives from informal salaried work into non-salaried work when mobility is low (most strongly for low-skilled men). Work permits (4) raise native informal salaried employment and (5) raise total formal employment in the economy when substitutability is high.

Predictions (1), (4), and (5) follow directly from Propositions 1, 4, and 5. They describe the effect of each shock on the market it directly hits and hold for any degree of worker mobility. Predictions (2) and (3) describe the migration shock's effect on the other two native margins, formal salaried and non-salaried employment, and their signs depend on how freely displaced natives re-sort across sectors. Both hold under limited mobility: formal salaried employment falls and non-salaried employment rises. Sufficient mobility reverses the formal-employment response, as natives re-sort into formal work, so we use its estimated sign to locate the economy on the mobility spectrum. The non-salaried response corroborates the mechanism but plays no role in this test. Its direction under limited mobility, natives moving into self-employment, is robust to modeling assumptions. Whether it reverses under high mobility is not, so we do not rely on the reversal.[26] Direct evidence that roughly 3 percent of informal Colombian workers transition to formal employment in a given year leads us to expect the low-mobility signs (Mondragón-Vélez et al., 2010).[27]

None of the five predictions concerns the effect of work permits on native formal employment: theory does not sign it, and its sign reveals whether migrant entry or the induced demand for formal labor dominates (Section 3.3). The two shocks also discipline each other through the model: how strongly natives' informal employment responds to the migration

[26] The asymmetry is intuitive. The formal-employment reversal pits the demand-side incidence of the shock against a single native margin, flight into formal work, and both are common to any model in which formal work is a distinct destination, so the threshold is robust. The non-salaried reversal pits two native margins against each other, the flow into self-employment and the flight into formal work, and which dominates turns on the strength of the flow into self-employment. Appendix E.13 re-derives the model with non-salaried work as a distinct occupation and shows that all five predictions and the direction of the non-salaried response hold in both formulations.

[27] We rely on external transition estimates because the public GEIH cannot be linked into an individual panel, so workers cannot be followed over time.

shock determines which force should dominate after work permits, and Section 5 shows that the estimated responses to the two shocks are mutually consistent. Sections 4 and 5 provide empirical tests of predictions (1) to (3) and (4), respectively. Data limitations prevent a direct test of prediction (5), so Section 5.4 quantifies it with the calibrated model at zero mobility.

4 The Effects of Informal Labor

4.1 Constructing the distance instrument

We construct our distance exposure measure Z_r following the established approach in the forced migration literature (Angrist and Kugler, 2003). For each of the 24 Colombian departments r, we calculate the migrant-weighted average inverse distance to the 23 Venezuelan states s, $Z_r = \sum_{s=1}^{23} \pi_s / d_{r,s}$, where $d_{r,s}$ represents travel distance between department and state capitals, and π_s denotes the share of Venezuelan migrants originating in state s, taken from the administrative registry of Venezuelan migrants (RAMV).[28] We standardize Z_r to have zero mean and unit variance, ensuring economically interpretable coefficients in our reduced-form specifications.

To evaluate the instrument's predictive power, we estimate an event study specification relating the migrant-to-native ratio $R_{r,t}$ to our distance exposure:

$$R_{r,t} = \sum_{j \neq 2013} \lambda_j (\text{year}_j \times Z_r) + f_r + f_t + \epsilon_{r,t} \tag{7}$$

where year dummies interact with distance exposure, and f_r and f_t represent department and year fixed effects, respectively.

Panel (a) of Figure 4 presents the estimated coefficients λ_j from equation 7, alongside the debiased first stage we introduce below. Distance exposure strongly predicts Venezuelan settlement in every post-treatment period, with a joint F-statistic of 278 for 2016–2019. The treatment intensity increases monotonically after 2015, reaching 2 percentage points by 2018 for a one standard deviation increase in distance exposure.[29]

The treatment ratio $R_{r,t}$ uses birthplace, not prior residence, to identify migrants: the numerator counts individuals born in Venezuela, while the denominator counts all individuals

[28] The weights π_s are national: they are the same for every Colombian department, so all cross-department variation in Z_r comes from travel distance alone. Weighting instead by the population of each Venezuelan state gives an instrument correlated 0.99 with this one across the 24 departments.

[29] The GEIH identifies Venezuelans and Colombians separately only from 2013. Before then, we hold each department's migrant-to-native ratio at its 2013 level, which is safe since in 2014 only 0.16% of the working-age population were Venezuelan.

born in Colombia. Colombian return-migrants, individuals born in Colombia who previously lived in Venezuela, are therefore excluded from the numerator by construction. We refer to $R_{r,t}$ throughout as the migrant-to-native ratio. Appendix Figure H.9 shows that this exclusion is inconsequential. Plotted on a common scale, return-migrant settlement responds to distance only through a small early uptake that plateaus after 2017. Venezuelan-born settlement rises steadily and grows several times larger. Our identifying variation is this post-2017 buildup, which return migration does not track.

We complete our shift-share instrument by interacting the share Z_r with the shift S_t: $Z_{r,t} = Z_r \times S_t$, where S_t is the national migration shift in year t, measured as the sum of the migrant-to-native ratio across the 24 departments.

4.2 The pre-trend problem

To visually and flexibly assess the pattern of outcomes $y_{r,t}$ captured by distance exposure relative to the beginning of the refugee crisis, we estimate an event study design. The basic nonparametric event study specification takes the form

$$y_{r,t} = \sum_{j \neq 2015} \theta_j (\text{year}_j \times Z_r) + f_r + f_t + \epsilon_{r,t}, \tag{8}$$

where the distance share Z_r is interacted with year dummies year_j, and f_r and f_t are department and year fixed effects. The coefficients θ_j compare the outcomes in departments close to and away from the border. Causal interpretation requires that, absent treatment, these departments would follow parallel trajectories. However, our event study estimates reveal systematic pre-treatment differences in trends across multiple labor market outcomes.

Specifically, departments closer to the Venezuelan border experienced declining salaried employment rates relative to interior departments throughout 2007–2015, creating negative trends correlated with our instrument. Visual inspection across various outcomes suggests that no single parametric correction (such as linear time trends) adequately addresses this pervasive pattern. Appendix H.1 documents why: the pre-trend reflects a distance-correlated macroeconomic slowdown that began before migration and breaks at its onset, so that neither year fixed effects nor a linear-trend adjustment can remove it. These findings raise fundamental concerns about the validity of distance-based instruments widely employed in the Venezuelan migration literature (Caruso et al., 2021; Delgado-Prieto, 2024; García-Suaza et al., 2024). To make progress, we use SIV.

4.3 The SIV solution

The SIV methodology combines synthetic control (SC) methods with instrumental variables estimation in two steps. First, we solve the standard synthetic control problem on pre-treatment outcomes and use the resulting weights to construct synthetic versions of the outcome, the treatment, and the instrument. Subtracting each synthetic series from its observed counterpart yields the debiased variables $\tilde{y}_{rt} = y_{rt} - \hat{y}_{rt}^{SC}$, $\tilde{R}_{rt} = R_{rt} - \hat{R}_{rt}^{SC}$, and $\tilde{Z}_{rt} = Z_{rt} - \hat{Z}_{rt}^{SC}$. Second, we estimate all specifications on the debiased data. Intuitively, the SC step eliminates systematic pre-treatment differences between border and interior departments, while the IV step continues to address migrants' endogenous location choices. Appendix H provides technical details of the algorithm.

In Section 4.5, we show the reduced-form results by estimating equation 8 separately using IV and SIV, with the only difference being that the instrument and the outcome are their debiased versions in SIV. To show economically meaningful results, we also present 2SLS estimates of the following system of equations using the debiased data:

$$\begin{aligned} \text{LATE:} \quad \tilde{y}_{r,t} &= \beta^{LATE} \tilde{R}_{r,t} + f_r + f_t + \epsilon_{r,t} \\ \tilde{R}_{r,t} &= \theta \tilde{Z}_{r,t} + g_r + g_t + \eta_{r,t} \end{aligned} \tag{9}$$

4.4 Threats to identification

Our empirical strategy compares departments close to the border with those further away. There are three main reasons why this comparison may not be valid.

First, border departments may experience differential impacts from Venezuela's economic crisis through trade relationships rather than migration. Appendix Figure C.4 documents that trade flow changes show no systematic correlation with our distance instrument, supporting the exclusion restriction.

Second, the Stable Unit Treatment Value Assumption (SUTVA) could be violated if labor or capital mobility generates spillover effects across departments. Empirically, Appendix Figure C.5 shows border departments experienced modest increases in out-migration and decreases in in-migration, but these effects are economically small and would only marginally bias our estimates toward zero. For capital spillovers, Gulek (2025) demonstrates that immigration-induced trade effects matter primarily for economically developed cities. Since Venezuelan migrants concentrate in less-developed border departments, capital reallocation concerns appear minimal.

Third, Venezuelan migration reduces native salaried employment, which changes the composition of the workers whose wages we observe. Our employment outcomes are pop-

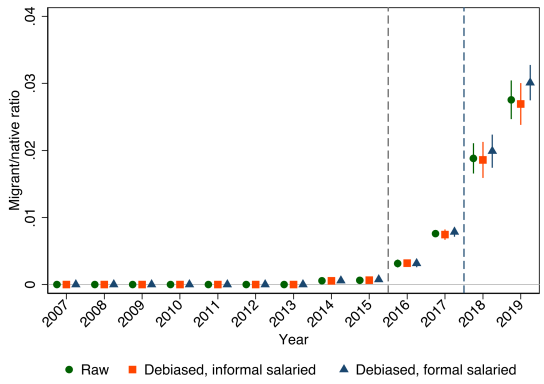
ulation shares within demographic cells, so selection leaves them untouched: the informal salaried employment rate falls whether the least or the most productive informal worker leaves. Wages are measured conditional on employment, so we report them as suggestive evidence (Appendix Figure C.2). We also measure the shift directly with the same SIV design: we estimate the effect of the migration shock on the mean age, years of education and female share of natives who remain in each sector (Appendix Figure C.7). The formal salaried workforce is stable once pre-existing trends are removed. The informal salaried workforce becomes modestly more educated and more male, the direction that displacement of the most marginal workers would produce.

4.5 Empirical Results

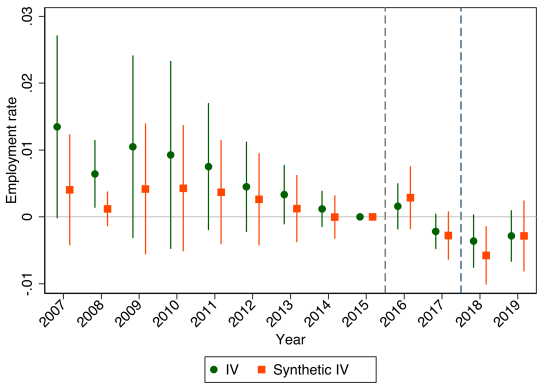
Figure 4 compares event study estimates using standard IV versus SIV methodologies for native salaried employment rates in informal (panel (b)) and formal (panel (c)) sectors.[30]

[30] Appendix H.1 explains why the two estimators differ where they do: for those outcomes the IV coefficients ride a macro-driven, distance-correlated pre-trend that SIV removes but a linear-trend adjustment cannot.

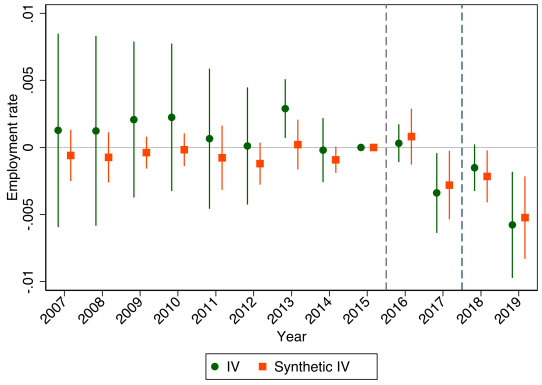
Figure 4: The First Stage and the Effects on Native Salaried Employment



(a) First stage, raw and debiased



(b) Informal Employment



(c) Formal Employment

Notes: Panel (a) plots the first stage, $R_{r,t} = \sum_{j \neq 2013} \lambda_j (\text{year}_j \times Z_r) + f_r + f_t + \eta_{r,t}$, estimated on the raw data and on the debiased data underlying panels (b) and (c). The instrument Z_r is standardized to mean zero and unit standard deviation. The base year is 2013, the first year in which the GEIH migration module identifies Venezuelan-born residents, so the migrant-to-native ratio is constant by construction before then and the 2007 to 2012 coefficients are mechanically zero. Migration remains negligible through 2015: the 2014 and 2015 coefficients are about 0.0006, roughly two percent of the 2019 value. The joint F-statistic for 2016 to 2019 is 278 on the raw data, 119 on the data debiased for informal salaried employment, and 155 for formal salaried employment, so the debiasing step leaves the instrument strong. Panels (b) and (c) plot the reduced-form event study $y_{r,t} = \sum_{j \neq 2015} \theta_j (\text{year}_j \times Z_r) + f_r + f_t + \epsilon_{r,t}$, with IV estimated on the raw data and SIV on the debiased data, as described in the text. Vertical dashed lines mark the onset of mass migration (late 2015) and the first PEP wave (2017q3). Bars are 95% confidence intervals. Standard errors are clustered at the department level. $N = 312$ (24 departments $\times$ 13 years) in every panel.

In panel (b), the IV estimates exhibit a pronounced negative trend between 2007–2014, indicating that border departments experienced systematically declining informal employment rates relative to interior departments well before Venezuelan migration began. This pre-trend likely violates the parallel trends assumption and suggests our instrument correlates with unobserved confounders affecting border department labor markets. The SIV

estimates eliminate this problematic pre-trend, confirming effective synthetic control matching in the pre-treatment period. Post-treatment estimates reveal substantial declines in native informal salaried employment during 2017–2019.

Note that in 2017 and 2018, the SIV estimates are larger in absolute terms than the IV estimates. This means that, absent the migrant treatment, informal salaried employment in departments close to the border would have *risen* relative to other departments. Put differently, the confounding trend not only slows down but flips sign in the post-period. This reversal is why SIV is the appropriate tool here. Any parametric correction, such as a department-specific linear trend, would extrapolate the negative 2007–2014 trajectory into the post-period and misstate the counterfactual in the opposite direction of the bias it was meant to fix. The synthetic controls instead track the donor departments year by year, so the debiased counterfactual absorbs the reversal as it happens.[31]

In panel (c), we observe formal native salaried employment losses in the post-period. Since the instrument does not predict systematic differences between border and non-border departments in the pre-period, relaxing the share exogeneity assumption via SIV does not change our conclusions.

A comparison of IV and SIV LATE estimates reveals that the two estimators converge when the full 2007–2019 sample is used but diverge meaningfully when the sample is truncated at 2017, leaving only two post-migration years (Appendix Figures H.13 and H.14). This pattern is consistent with pre-trend bias dominating the IV when the post-period is short, and the treatment effect signal overwhelming the bias as the post-period grows, precisely the scenario the SIV is designed to address.

These patterns match key theoretical predictions. Before work permits became widely available in 2018, Venezuelan migration represented a predominantly informal labor supply shock. The observed decline in native informal salaried employment matches prediction (1) from our theoretical framework. Simultaneously, the decline in native formal salaried employment matches prediction (2), indicating that informal and formal workers are gross substitutes in production and that natives' mobility across sectors is limited.

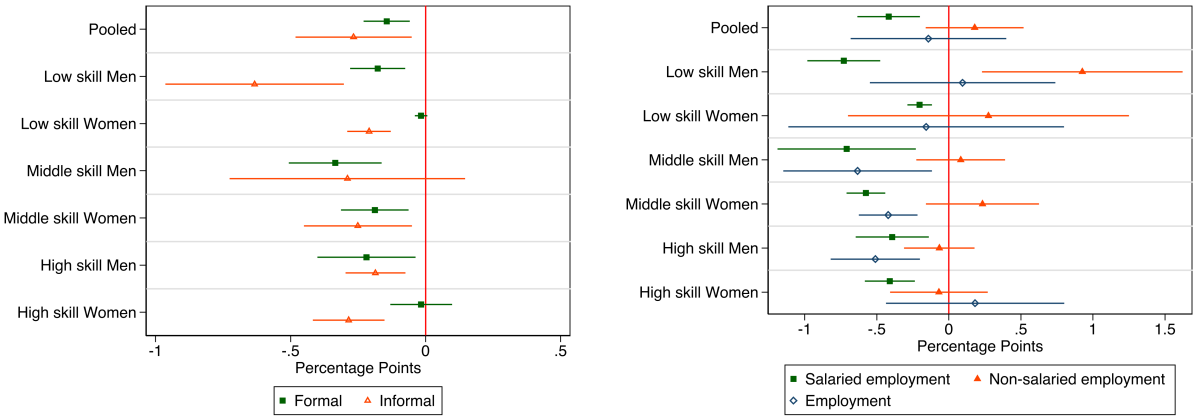
Figure 5 presents 2SLS estimates from equation 9 across different worker categories and skill groups. Panel A compares informal versus formal salaried employment effects, while Panel B examines salaried versus non-salaried employment patterns. Appendix Table C.1 reports all of the 2SLS estimates shown in this section.

The first row of Figure 5a shows that the mostly informal immigration shock caused

[31] A reversal of the confounding trend is plausible: Colombian GDP per capita rose from $4,770 in 2007 to $8,187 in 2014, then fell to $5,960 in 2016. If the more developed interior departments spearheaded the boom, the 2015 slowdown would compress the very gap that generated the negative pre-trend.

significant declines in the informal and formal native salaried employment rates. A 1 pp increase in the migrant-to-native ratio decreases the native salaried employment rate by 0.27 pp in the informal sector and 0.14 pp in the formal sector. These effects are consistent with predictions (1) and (2) and imply that formal and informal labor are highly substitutable in production. To contextualize these magnitudes, consider a municipality with 1,000 working-age natives. An influx of 10 Venezuelan migrants (representing a 1 percentage point increase in the migrant-to-native ratio) would displace 2.7 informal salaried workers and 1.4 formal salaried workers. Given that Venezuelan migrants maintain a 68% employment rate, this implies that 6.8 new migrant workers displace 4.1 native salaried positions, suggesting substantial displacement.

Figure 5: LATE estimates



(a) Informal vs Formal

(b) Salaried vs Non-salaried

Notes: The estimates are 2SLS coefficients from equation 9 and capture the effect of the migration shock (the migrant-to-native ratio, instrumented by debiased distance exposure) on native employment rates, estimated on the 2007–2019 sample; they predate and are distinct from the work permit effects of Section 5. Panel A shows the estimated effects on the informal and formal salaried employment rates, and Panel B on salaried, non-salaried, and total employment rates; the first row pools all demographic groups, and the remaining rows estimate each skill-sex group separately (outcomes are employment shares of the group's own working-age population). Bars are 95% confidence intervals. Standard errors are clustered at the department level. $N = 312$ (24 departments $\times$ 13 years) per outcome.

Rows 2–7 of Figure 5a reveal that employment losses span virtually all education-gender categories, with both informal and formal salaried employment declining across skill groups. This broad-based pattern provides reassurance that our results reflect genuine labor market competition rather than skill-specific demand shocks. Importantly, since only middle- and high-skilled male migrants typically accessed formal employment through work permits, the widespread formal employment losses among native groups support our theoretical mecha-

nism emphasizing cross-sector substitutability.[32]

Panel B of Figure 5 tests prediction (3), that natives displaced from salaried work transition into non-salaried employment, and the estimates corroborate that mechanism.[33] As Section 3.4 notes, they play no part in locating the economy on the mobility spectrum, which rests on the formal-employment sign of prediction (2). The direction of the non-salaried response under limited mobility is common to both formulations of that margin; only its reversal at high mobility is specific to the hours model (Appendix E.13). While salaried employment declines significantly across most skill groups, non-salaried employment increases, particularly among low-skilled workers. Although the non-salaried employment increases lack statistical significance in the aggregate, they are economically meaningful and concentrated among demographic groups predicted by our theoretical framework. In the pooled sample total employment shows no significant change (Appendix Table C.1). For low-skill men the two margins offset, so their outside option to salaried employment is non-salaried work rather than non-employment. Decomposing the non-salaried margin for low-skill men shows that the transition runs almost entirely through own-account self-employment, 0.76 pp of the 0.93 pp total effect (Appendix Figure C.8).

Our empirical estimates imply that informal and formal labor are gross substitutes in production. To provide a parameter estimate, we estimate the model using our SIV estimates and moments from the data. Appendix F details the estimation. We estimate $\rho \approx 0.90$, which corresponds to a substitution elasticity $\sigma = \frac{1}{1-\rho}$ of approximately 10. This substantial elasticity aligns with Colombia's institutional context, where informal work frequently occurs within the same industries and sometimes the same enterprises as formal employment. The estimate is similar in magnitude to the elasticity of substitution in Turkey (Gulek, 2023) and indicates that formal and informal labor are strong gross substitutes, close to the near-perfect-substitutability benchmark commonly employed in recent structural models of informal labor markets (Ulyssea, 2018, 2020).

[32] In percentage terms, low-skill men lose the most formal salaried jobs while high-skill natives lose the least, consistent with a binding minimum wage in the low-skill market causing demand adjustments on the employment margin rather than the wage margin.

[33] Panel B pools formal and informal work within both categories. This does not blur the comparison with prediction (3), which concerns the informal margin, because non-salaried work is overwhelmingly informal: 91% of non-salaried natives lack social security coverage, against 38% of private salaried workers. Among low-skill men, the group with the largest non-salaried response, the figure is 95%. The non-salaried outcome is therefore, in practice, informal non-salaried work. Panel A splits the salaried category by formality, and both of its components decline.

4.6 Robustness Checks

If contemporaneous industry-specific demand shocks drove our results rather than immigration, the losses would concentrate in a few industries. They do not. Estimating equation 9 by major industry, we find salaried employment falling in nearly every one, the exception being Health and Social Services. Displaced natives move into non-salaried work, most strongly in Manufacturing (Appendix Figure C.3).

SIV's reliability rests on three diagnostics, which Appendix H.2 reports in full. First, the debiased first stage stays strong. Because SIV builds outcome-specific synthetic controls, each specification has its own debiased instrument and treatment. Panel (a) of Figure 4 shows the debiased first stage for our two main outcomes. Across the 35 main specifications the minimum F-statistic exceeds 46, and more than three quarters exceed 100. Second, the synthetic controls fit the training period well. Our conclusions hold when the synthetic controls are computed on pre-treatment demeaned data, which adds a constant shift to the fit (Doudchenko and Imbens, 2016; Abadie and Vives-i Bastida, 2022). Third, shortening the training window from 2007–2015 to 2007–2012 leaves the estimates stable, so the nine-year window is not overfitting.

We also replicate the main results with an alternative instrument, the 2005 census Venezuelan-born population share across departments (enclave IV). It has a weaker first stage than the distance instrument. Debiased the same way, it delivers qualitatively similar LATE estimates with wider confidence intervals (Appendix Figures H.10–H.12).

5 The Effects of Work Permits

5.1 Identification

The identification challenge is that the spatial distribution of work permits strongly correlates with the migration shock. PEPs were distributed through registration at migration offices, so border departments that received more migrants also processed proportionally more permits (Appendix Figure B.1). Once migration intensity is controlled for, insufficient spatial variation remains to identify work permit effects using standard approaches.

To make progress, we exploit the fact that work permits moved only some migrants into formal employment. As Figure 2 shows, formal employment rises sharply among high-skill Venezuelan men after PEP-1 and among middle-skill men after PEP-3, but stays flat for low-skill men and for women across all three waves.[34] Native workers in the flat segments

[34]The confidence intervals in Figure 2 confirm these patterns: the post-PEP increases for high- and middle-skill men are significant, while the control series are indistinguishable from their pre-PEP baselines.

therefore faced no new formal competition from the permit waves, which makes them natural comparison groups.

For PEP waves 1 and 2, this absence of a response is institutional rather than endogenous. These waves required applicants to hold a valid passport and a legally recognized migratory status. During 2016–2018, obtaining a Venezuelan passport was prohibitively costly: bureaucratic collapse and corruption raised the effective cost to $500 or more in bribes, at a time when hyperinflation had collapsed Venezuelan incomes (Rochlin, 2025; Chaves-González and Delgado, 2023). This created a strong wealth screen that disproportionately excluded lower-skilled migrants, who were far less likely to possess formal documentation upon arrival in Colombia.[35] The fact that our main results hold when estimated using only the variation induced by PEP waves 1 and 2, where low-skill non-participation is driven by institutional exclusion rather than labor market sorting, provides strong support for the validity of our control group.[36]

PEP-3 relaxed documentation requirements through the RAMV registration, making low-skill migrants eligible in principle. Figure 2 nonetheless shows that they did not transition to formal employment. Firms may have chosen to keep them in informal positions. Our model also rationalizes the low take-up. With a single permit cost calibrated to the aggregate formalization rate, it delivers skill-graded formalization endogenously: high-skill migrants formalize at roughly four times the rate of low-skill migrants, because the formal-informal return gap widens with skill (Appendix E.9). For our identification the reason does not matter. The triple difference compares native groups whose closest migrant substitutes did and did not formalize. If low-skill migrants did not enter formal employment, low-skill natives were not exposed to increased formal-sector competition from migrants and are valid controls.[37]

Moreover, high- and middle-skill men were impacted by the work permits at different

We treat all women as untreated because neither high- nor middle-skill women formalize in any sustained way, and Section 5.3 shows our estimates are insensitive to that choice.

[35] Bahar et al. (2021) document that the RAMV census, which by construction registered only undocumented Venezuelans, identified over 440,000 migrants who were ineligible for PEP waves 1 and 2 precisely because they lacked proper documentation.

[36] The high-skill-men treatment is identified entirely from PEP waves 1 and 2; the high-skill-men estimates appear in the event study of Appendix Figure D.2 and the group-level building block of Appendix Figure D.4, both of which reproduce the baseline signs.

[37] A remaining threat would be general equilibrium spillovers from high-skill migrant formalization to low-skill native outcomes, violating the stable unit treatment value assumption across skill groups. Our department-by-quarter fixed effects absorb any shock common to all skill groups within a department, and our skill-group-by-quarter fixed effects absorb any shock common across departments within a skill group, so the triple difference identifies the effect on treated groups relative to control groups rather than the economy-wide level. The component that scales with local migrant exposure is specific to both a skill group and a department, and is not absorbed. Recovering the economy-wide response would require a structural model with multiple worker types, which we do not build.

times: PEP-1 and PEP-2 enabled high-skill Venezuelans to begin applying in the third quarter of 2017 and to formalize over the following quarters, while PEP-3 enabled middle-skill men to find formal jobs starting in 2019q1. This variation across skill cells provides a third dimension in which the work permit treatment intensity varies in addition to space and time. It allows for a triple DiD design that identifies the effect of work permits by partialing out the effects of immigration.

The triple differences in differences design

Our identification strategy rests on the premise that middle- and high-skilled native men face greater exposure to work permit effects than other demographic groups, since they are closest substitutes to the skilled migrant men who could utilize work permits effectively. We formalize this logic in a triple difference-in-differences framework that compares outcomes across skill groups, time periods, and departments.

We define treatment exposure as:

$$D_{brt} = \begin{cases} 1 & \text{if } b = \text{high-skill men } \& \ t > 2017q3 \ \& \ \text{department} = \text{close to border} \\ 1 & \text{if } b = \text{middle-skill men } \& \ t > 2018q4 \ \& \ \text{department} = \text{close to border} \\ 0 & \text{otherwise} \end{cases} \tag{10}$$

where treatment switches on when work permits become available to each skill group in border-proximate departments; "close to border" designates the three departments with the highest distance exposure Z_r: La Guajira, Norte de Santander, and Cesar. We use a binary treatment indicator for interpretational clarity.

Formally, we estimate the following regression equation:

$$Y_{brt} = \beta D_{brt} + f_{br} + f_{bt} + f_{rt} + \epsilon_{brt}, \tag{11}$$

where Y_{brt} represents labor market outcomes for demographic group b in department r at time t. The fixed effects structure is crucial for identification: f_{br} (group-department effects) control for time-invariant differences in how departments affect specific demographic groups; f_{bt} (group-time effects) capture economy-wide trends affecting particular skill-gender categories; and f_{rt} (department-time effects) absorb all department-level shocks, including ongoing migration impacts.

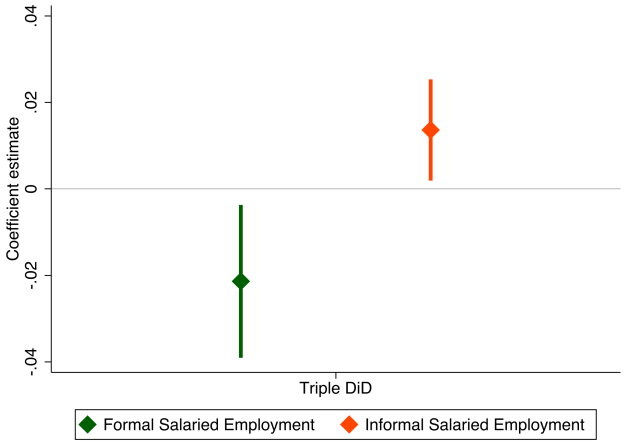
The identification strategy here differs from Section 4.5 in how it handles the differential trends between border and interior departments. In the shift-share analysis, we showed that border distance correlates with pre-existing trends in labor market outcomes, motivating the

Synthetic IV correction. These trends operate at the department level: border departments experienced declining salaried employment rates relative to the interior throughout 2007–2015, regardless of skill group. In the triple difference-in-differences, the department × time fixed effects f_{rt} absorb these department-level trajectories, since they control for all time-varying differences between border and interior departments. Consequently, the identifying variation in equation 11 comes from within-department comparisons across skill groups: whether treated skill-sex categories experienced differential employment changes relative to control categories in the same department and quarter. This assumption, parallel trends across skill groups within departments, is weaker than the cross-department parallel trends assumption required in Section 4.5.

5.2 Empirical results

Figure 6 presents our central findings from estimating equation 11 for informal and formal salaried employment rates. The results confirm prediction (4) of our theoretical framework and answer the question it deliberately leaves open, the effect of work permits on native formal employment.

Figure 6: Triple DiD Estimate



Notes: The figure shows estimated coefficients from equation 11 where the dependent variables are informal and formal salaried employment rates. Treatment exposure is defined as a binary indicator for high-skilled men after 2017q3 and middle-skilled men after 2018q4 in the three most exposed departments (La Guajira, Norte de Santander, and Cesar). Standard errors are clustered at the skill-group × department level. Confidence intervals show 95% confidence levels. Estimation window: 2017q1–2019q4; N = 1,728 (24 departments × 6 skill-sex groups × 12 quarters).

Work permits generate a statistically significant increase in native informal salaried employment and a decrease in native formal salaried employment. In magnitudes, the intent-

to-treat coefficients are +1.4 percentage points for informal and −2.1 percentage points for formal salaried employment; relative to the treated groups' baseline rates of roughly 10 and 28 percent (Appendix Table A.1), these correspond to an increase of about 14 percent and a decline of about 8 percent for exposed groups in the most exposed departments. The first result confirms prediction (4). The second answers the question that theory left open in Section 3: whether work permits create losers among native workers. In principle, permits could have hurt no native worker, with informal natives gaining from reduced competition and formal natives gaining from firms' increased demand for formal labor. The data show that Colombia is not in that scenario: migrant entry into the formal sector dominates the induced demand response, and permits redistribute between native workers, benefiting informal natives at the expense of formal natives.

5.3 Robustness Checks

Our results remain robust to expanding the pre-period observations (Appendix Figure D.1a) or to using alternative cutoffs (Appendix Figure D.1b). Widening the exposed set to the four most exposed departments gives −1.9 percentage points for formal and +1.0 for informal salaried employment. The continuous distance exposure Z_r gives the same signs, but the formal estimate is significant only at the 10 percent level (Appendix Figure D.1c). We also provide the implied event-study estimates of the triple DiD design (Appendix Figure D.2). Joint F-tests of the pre-period coefficients do not reject for middle-skill men's formal salaried employment ($p = 0.357$), the cell in which our formal-employment result is most precisely estimated. They do reject for the other three: high-skill men's formal ($p = 0.004$) and informal ($p = 0.024$) employment, and middle-skill men's informal employment ($p < 0.001$). The rejections reflect quarter-to-quarter volatility rather than drift: the individual pre-period coefficients are almost never significant, and a linear differential trend fitted to the pre-period is indistinguishable from zero in all four cells (Appendix D).

We do not claim that every cell is clean. The cell that carries our formal-employment result is also the cleanest: none of its pre-period coefficients is individually significant. Where the joint tests reject, we rely on three features of the design rather than on the tests. First, the individual pre-period coefficients are almost never significant. Second, no cell shows a fitted pre-trend. Third, the two treated groups are treated five quarters apart, so a confounder would have to hit high-skill and middle-skill men in exposed departments at those two dates.

Our estimates do not depend on how high-skill women are treated. They show a modest rise in formal employment from 2018q3 onward, but it never clears the 1% threshold and survives no multiple-testing correction (Figure 2), so we treat them as untreated. We re-

estimate the triple difference under four treatments of this group, in three designs: the staggered baseline, high-skill men alone, and middle-skill men alone. The direct test is to drop high-skill women from the sample, which removes them from the treatment group and the control group alike. The middle-skill-men estimate is unchanged at -2.5 percentage points ($p = 0.018$), and every informal estimate is unchanged. In the staggered design the formal coefficient moves from -2.1 to -1.8 percentage points and no longer clears 5 percent. Two of the three alternatives lose individual significance, for mechanical reasons. Restricting to men halves the sample, and we do not claim that the men-only formal estimate stands on its own. Moving high-skill women into the treatment group averages our estimate with their own positive coefficient. A pairs cluster bootstrap on 10,000 replications cannot distinguish any alternative from its own baseline at the 5 percent level. Appendix Table D.1 and Figure D.5 report the results.

Estimated separately, the two treated groups replicate one another. Formal salaried employment falls by 2.4 percentage points among high-skill men and by 2.5 percentage points among middle-skill men, and informal salaried employment rises by 0.7 and 1.7 percentage points respectively. The two groups are treated five quarters apart and on disjoint treatment variation, so their agreement is a replication rather than a restatement. Neither group is fully powered on its own: the middle-skill-men formal estimate is significant ($p = 0.011$) and is unchanged when high-skill women are dropped ($p = 0.018$), while the high-skill-men estimate of the same magnitude carries a wider standard error ($p = 0.071$). The staggered design pools the two, which is what it is for.

Finally, we decompose the triple DiD into its building blocks, once from department-level DiDs and once from group-level DiDs. Neither decomposition reveals an outlier: the border-interior gap does not come from a single department, and the treated groups differ from every control group. Appendix D reports both exercises.

5.4 Data limitations and model-based quantification

We test predictions (1) to (4) directly, but data limitations prevent a direct test of prediction (5): work permits raise total formal employment when substitutability is high. The GEIH survey lacks sufficient precision to measure Venezuelan employment across department-skill-time cells, while administrative records of PEP holders cannot be disaggregated by skill categories.[38]

[38]The empirical estimates compare departments, while the model describes a single labor market. Internal migration responses to the shock are negligible and migrants did not concentrate in one of the country's main economic regions, so the estimates are unlikely to mask significant general equilibrium effects, and we use them to estimate the model.

Before using the model to quantify this final mechanism, we ask whether it predicts what the triple DiD found when its parameters are estimated from the migration shock alone. Section 3 left the effect of work permits on native formal employment unsigned: natives lose formal jobs if migrant entry into the formal sector dominates the increase in formal labor demand induced by the higher informal wage, and gain otherwise. The estimated model takes a stand, because the migration shock disciplines both forces in this comparison. The estimation implies aggregate demand elasticities of -0.65 for informal labor and 1.32 for formal labor with respect to the informal wage. The migration estimates then pin down how strongly native informal employment responds to wages: natives offset 87 percent of migrant entry into informal salaried work, which implies a native informal labor supply elasticity of approximately 4.3.[39] With supply this elastic, the informal market absorbs the permit-induced migrant exodus mostly through employment rather than wages: the informal wage rises too little for the induced demand response to overturn migrant entry in the formal market.[40] The estimated model therefore predicts that the formal wage falls and native formal employment declines, which is what the triple DiD finds. We read this as a consistency result rather than a sharp test (Appendix E reports the margins). The model, estimated on one natural experiment, predicts the sign pattern of a second, independent one.

Given the model parameters and the exact number of work permits provided, we can point estimate the policy's effect on formal job creation in three steps. First, the GEIH gives the number of Venezuelans who moved into formal jobs with the new permits, and hence the fall in migrant informal labor supply. Second, the informal labor demand curve and the native supply response estimated from the migration shock give the resulting rise in the informal wage. Third, the formal labor demand elasticity times this wage change gives the percentage change in formal labor demand, and hence the number of formal jobs created. Migración Colombia issued roughly 440,000 permits across the three waves in the departments our data cover (67,580 under PEP-1, 110,841 under PEP-2, and 260,994 under PEP-3), of which about 378,000 went to migrants of working age. We estimate that approximately 68,000 of those permit holders moved from informal to formal employment, an aggregate formalization rate of about 18 percent.[41] This in turn has created around 23,100 formal jobs in the economy.

[39] Natives lose 0.266 informal salaried jobs (Appendix Table C.1) per 0.307 migrant entrants, the private-salaried employment rate of working-age Venezuelans in 2019. The migration shock moves native informal salaried employment by the LATE estimate and the informal wage by the total supply change divided by the demand elasticity; the ratio of the two identifies the supply elasticity. The implied value is $0.65 \times 0.2665/(0.3067 - 0.2665) \approx 4.3$.

[40] Formally, at low mobility the formal wage falls if and only if $\nu(|\epsilon_{L_i,w_i}| + \epsilon_{is}) > \epsilon_{L_f,w_i}$, where ν is the ratio of informal to formal salaried employment and ϵ_{is} is the native informal labor supply elasticity. At the estimated and observed values, $0.68 \times (0.65 + 4.3) = 3.4 > 1.32$.

[41] The GEIH number is the rise in formally employed, private salaried Venezuelans from 2015 to 2019. We also validate this number using administrative records. By the end of 2019, 73,360 Venezuelans with

Assuming these jobs pay the 2019 mean monthly wage of formal salaried workers, taxed at the 32.6 percent effective payroll rate of Appendix F, the permits generate around 42 million USD in annual tax revenue.

We evaluate the counterfactual with the zero-mobility version of the model, the case our estimates support.[42]

Limited mobility may seem at odds with an elasticity of substitution near 10. The elasticity describes the production technology, and mobility describes workers' preferences. Appendix E shows that no theoretical restriction links the two and that the implied mobility is in line with evidence that transitions between informal and formal jobs are rare in Colombia (Mondragón-Vélez et al., 2010).

Work permits may also raise productivity by letting migrants work closer to their skills: a Venezuelan engineer who works informally in a restaurant for lack of work authorization can move into professional services once a permit arrives. Appendix G tests this channel with a quadruple difference in differences design that adds industry exposure to the triple difference above. A 10 percentage point increase in an industry's skill intensity raises the wages of more exposed natives by 0.87 percent. The same increase in formality raises them by 0.47 percent, significant only at the 10 percent level. For skill intensity the gain is as large at the 75th percentile of the wage distribution as at the 25th, which composition alone cannot generate. We cannot separate formality from skill intensity, and we cannot rule out industry-specific demand shocks that coincide with the permit waves, so we read this as suggestive evidence that permits generate gains beyond the jobs they create.

6 Conclusion

This paper analyzes how informal immigration and work permit policies jointly affect labor markets in developing economies. Using the Venezuelan refugee crisis in Colombia as a quasi-experiment, we develop an equilibrium model that incorporates both the formal-informal and the salaried-nonsalaried margins, and we implement two research designs that separately identify the effects of the migration shock and of the subsequent work permits.

Our findings demonstrate that predominantly informal immigration generates displacement across both informal and formal sectors. Displaced natives, especially low-skill men, move into non-salaried employment rather than unemployment (Appendix Figure C.6).

PEPs were paying into social security through their jobs. Using this number as opposed to 68,000 does not change our counterfactual estimates in a meaningful way.

[42] Allowing for worker mobility scales the counterfactual down by a closed-form factor (Appendix E). Over the mobility range consistent with the estimated sign of prediction (2), the job-creation figure retains at least 93 percent of its zero-mobility value (a band of roughly 21,500 to 23,100 jobs), so the choice matters little.

These results indicate a high substitutability between informal and formal workers (elasticity ≈ 10) and highlight the importance of non-salaried employment as an adjustment mechanism in the labor markets of developing countries, dynamics that would be missed by focusing solely on aggregate employment effects.

Work permits fundamentally alter these dynamics by enabling sectoral reallocation that generates net economic benefits. While permits reduce informal sector competition and increase formal sector competition, the resulting wage changes induce firms to substitute toward formal employment. In our setting, this creates approximately 23,100 net new formal jobs. We also find suggestive evidence, reported in Appendix G, that permits generate productivity spillovers by reducing skill mismatch.

The results carry policy weight as refugee populations continue growing globally while most lack work authorization. Our analysis suggests that work permits not only benefit migrants but can generate substantial returns for host countries through increased formalization, job creation, and productivity gains. Permits are only one integration tool, however. Access to education and healthcare may matter equally for long-run labor market integration, as may language training and credential recognition. Separating these forces is a natural direction for future theoretical and empirical work.

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